

1. Scientific Notation (form: $a \times 10^n$, where $1 \leq a < 10$)

- a. $0.00000000356 = 3.56 \times 10^{-9}$
- b. $934,000,000 = 9.34 \times 10^8$
- c. $847 = 8.47 \times 10^2$
- d. $0.00092 = 9.2 \times 10^{-4}$
- e. $3,510,000 = 3.51 \times 10^6$

2. Engineering Notation (exponent must be a multiple of 3)

- a. $0.00000000356 = 3.56 \times 10^{-9}$ (nano)
- b. $934,000,000 = 934 \times 10^6$ (mega)
- c. $847 = 847 \times 10^0$ (base)
- d. $0.00092 = 920 \times 10^{-6}$ (micro)
- e. $3,510,000 = 3.51 \times 10^6$ (mega)

3. SI Prefix Notation (keep the units)

- a. $0.000047 \text{ F} = 47 \mu\text{F}$
- b. $17,500,000 \text{ Hz} = 17.5 \text{ MHz}$
- c. $0.0000000157 \text{ A} = 15.7 \text{ nA}$
- d. $6,800,000 \Omega = 6.8 \text{ M}\Omega$
- e. $0.00425 \text{ V} = 4.25 \text{ mV}$

4. Convert to the Specified SI Prefix

- a. $6800 \text{ pF} = 0.0068 \mu\text{F}$
- b. $2.7 \text{ M}\Omega = 2700 \text{ k}\Omega$
- c. $4.24 \text{ GHz} = 4240 \text{ MHz}$
- d. $25.67 \mu\text{F} = 0.02567 \text{ mF}$
- e. $0.0127 \text{ nSec} = 12.7 \text{ pSec}$